Chemical Engineering Journal 523 (2025) 168425

Contents lists available at ScienceDirect

Chemical Engineering Journal

journal homepage: www.elsevier.com/locate/cej

Improving sludge solubilization and biogas upgrading with eco-friendly garbage enzyme and microaerobic pretreatment: Operational parameters and energy balance

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ARTICLE INFO

Keywords:
Garbage enzymes
Biogas upgrading
Solubilization
Energy balance
Anaerobic digestion
Microaerobic

ABSTRACT

A comprehensive study was performed on the effects of combined garbage enzyme (GE) fermented from orange peels and micro-aerobic (MA) to improve solubilization of sewage sludge (SS) and biogas production with a special focus on energy. The enhanced biodegradability index (sCOD) of SS was optimized under operational parameters including different volume ratios of GE (5, 10, and 15 % *v*/v), air flow rate (AFR) (0.1–0.4 VVM) and time (10–40 h). The effects of combined GE and MA pretreatment on the operational parameters and biomethane potential (BMP) was performed under mesophilic temperature (37 ± 1 °C) in a batch mode reactor. The results indicated that the highest amylase activity (80.08 U/mL), protease activity (0.94 U/mL), lipase activity (44 U/mL) were achieved from the fermentation of orange peels. An increase of sCOD (2477 mg/L) was achieved in R4, when simultaneous GE (10 % *V*/V) and MA pretreatment methods were used to improve SS solubilization. The highest BMP (609.46 mLCH_4/g VS removed) was achieved in R4, which was 1.76 fold higher than the control. The energy analysis indicated that the highest energy yield (85.45 kJ/g VS) was achieved in anaerobic bioreactor assissted with simultaneous GE and MA pretreatments. The present study is unique in the sense that it meets the criteria of UNSDGs Program #SGD6, #SDG7, #SDG12 and #SGD13.

1. Introduction

Over the recent years, explosive population growth led to the production of a large amount of wastewater [1]. The treatment of wastewater produces a huge volume of sludge and biosolids in domestic and industrial wastewater treatment plant (WWTP) [2]. In addition to harmful human and environmental effects, sludge and biosolids management account for approximately 50 % of the total cost for wastewater treatment [3–5]. Nowadays, a variety of physical, chemical and biological methods have been used to treat sludge in order to sludge reduction, sludge stabilization and value-added products [6]. It is important to note that physical and chemical treatment with different limitations including high cost and production of chemical sludge with high contamination hare rarely used in sludge management [7]. The biological processes which take advantages of low cost, high stabilization and energy production have grasped much attention over the recent decades [8]. Anaerobic digestion (AD), as a biological process, is known as the most common technology for sludge stabilization, biogas and energy production [9]. AD converts organic solids present in sludge in the absence of oxygen into final products such as methane, carbon dioxide and harmless substances [10,11]. Overall, AD with four sequential stages (hydrolysis, acidogenesis, acteogenesis acetogenesis and methanogenesis) convert the organic matter into final products [12]. Hydrolysis is known as the rate-limiting step in AD; the high-molecular

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https://doi.org/10.1016/j.cej.2025.168425
Received 23 June 2025; Received in revised form 10 September 2025; Accepted 11 September 2025
Available online 13 September 2025
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weight compounds are converted to smaller molecules that are consumed by the microorganisms [13]. A variety of different pretreatment methods including physical (thermal) [14] chemical (alkalinity addition) [15] and biological (enzyme and micronutrient addition) [16] have been employed in order to accelerate the hydrolysis stage in AD, consequently enhancing methane and energy production [17].

The addition of hydrolytic enzyme is one of the biological pretreatment methods to enhance the hydrolysis stage and biogas production from AD fed with recalcitrant waste and sewage sludge [18]. However, the main limitation imposed for the synthetic hydrolytic enzyme is the high cost [19]. Alternatively, the enzyme production from organic waste in the fermentation process, called "garbage enzyme"is considered to be an eco-friendly process, easy operation and low cost [20]. On the other hand, hydrolytic enzymes containing lipase, amylase and protease are easily secreted by microorganisms in the fermentation process of organic solid waste supplemented with carbon source (brown sugar) and wastewater (with consortium of microorganisms) [21]. In recent years, GE have grasped much attention for different environmental applications such as soil remediation, wastewater treatment, landfill leachate treatment and composting [22].

Negi et al., [20] examined the vermicomposting process supplemented with GE in order to degrade food waste along with wheat straw and biomass waste. The authors reported that GE led to higher quality of compost with higher rate of organic matter degradation (0.15 d^{-1}) compared to control 0.12 d^{-1}). Rani et al., [23] focused on lowering the landfill pollution index (LPI) of three landfills in Delhi, India using different volume ratios of GE enzyme (5, 10, 15 and 20 % (*v*/v)) within 28 days. The authors used 18 parameters to calculate LPI. The authors reported the maximum LPI reduction of 45.47 % when 20 % volume ratio of GE was added to the landfill leachate (LL).

The integration of different pretreatment methods has been proposed for synergetic effects of hydrolysis and solubilization of organic matters in the AD process [24]. Micro-aerobic (MA) is a biological pretreatment method, involving the application of a small dose of air or oxygen (usually between 0.1 and 1 mg of oxygen/ l reactor per day) into an anaerobic system [25]. MA can improve the hydrolysis and solubilization of organic matters as well as increasing the lipase activity. Rashvanlou et al. (2021) investigated the roles of MA pretreatment on sewage sludge solubilization and methane production in the AD process [26]. The results indicated that MA (volume ratio of 0.2 (vvm) and duration time: 40 h) led to increased sCOD from 5210 mg/L to 8580 mg/L with increased methane production by 221 %. In the prior literature, no comprehensive study has been made on the application of combined GE and MA for enhancing the sewage sludge solubilizationa and consequently improving the methane production in AD. Therefore, the present study was developed to investigate the roles of synergetic effects of GE and MA pretreatment on sludge solubilization, methane production and energy balance.

2. Methods and materials

2.1. Material characteristics

To investigate the effect of combined pretreatment methods (GE + MA) on biogas production in AD, primary sludge, secondary sludge and inoculum samples were withdrawn from the full-scale south Tehran (Iran) wastewater treatment plant (STWWTP). The digested sludge as output of anaerobic digestion (AD) were prepared for inoculum in AD reactors and all biological samples were immediately transferred to the laboratory at 4 °C for further analysis. The physicochemical characteristics of primary sludge, secondary sludge and inoculum were determined as per procedures outlined in the standard method for examination of water and wastewater [24]. Table 1 summarizes physicochemical characteristics of primary sludge, secondary sludge and inoculum samples used in the study.

2.2. Production of garbage enzyme

In this study, orange fruit peels used most commonly in Iran during autumn and winter seasons were employed for the GE production. The orange peels were collected from the fruit markets and stored in a refrigerator at 4 °C for further analysis. Then, the crushed orange peels, brown sugar (as carbon source) and wastewater were mixed in the ratio of 1:3:10 (*w*/*v*) and taken in 500-mL bottles under anaerobic situation for fermentation process. Wastewater samples were taken from WWTP of food company. To ensure anaerobic condition, nitrogen gas was purged into different reactors for 5 min and the completely mixed fermentative reactors were incubated at 37 °C ± 1 for 21 days. After 3 weeks of fermentation, to measure the GE produced from the fermentation process, light yellowish-brown liquid supernatant were centrifuged at 3000 rpm for 30 min for further analysis [27].

2.3. GE (amylase, protease and lipase) measurement

2.3.1. Amylase activity

The amylase activity in GE were determined based as per Bernfeld's instructions, 3,5-dinitrosalicylic acid (DNSA) method and 1 % starch as a substrate. Then the GE samples were centrifuged to separate supernatants for further analysis [28]. According to Bernfeld's instructions, 0.5 mL of GE sample was mixed with 0.5 mL of 1 % starch solution and incubated at 37 °C for 15 min. Next, 1 mL of DNSA colored solution was added to the mixture and heated to 100 °C for 10 min. After cooling at room temperature, 5 mL of distilled water was added to the produced solution. All steps were repeated for control sample and distilled water without addition of starch addition. The absorbance of different GE samples were recorded at 540 nm against blank using spectrophotometry (Hach Company, DR-6000). Detailed information of the standard curve and color spectrum of different samples are provided in the supporting information (see Table A.1 and Fig. A.1). Next, the amount of maltose released from maltose and amylase activity (U/mg) were calculated as:

$$\text{Amylase activate (u/mL)} = \frac{(\text{ released maltose (mL)} \times \text{dilution factor })}{\text{Volume solution enzyme (mL)}} \quad (1)$$

2.3.2. Protease activity

The Bernfeld method was used to measure the protease activity in GE produced from the fermentation process [29]. At first, pH of the GE sample was initially adjusted to 7.5 by adding NaOH (1 M), then 2.5 mL of 1 % casein solution was combined with 1 mL of GE and the pH was re-

Table 1
Characteristics of sludge and inoculum.

Parameter	Inoculum	Primary sludge	Secondary sludge
pH	7.47 ± 0.1	6.14 ± 0.09	7.14 ± 0.08
TS (%)	2.96 ± 0.08	3.2 ± 0.06	5.2 ± 0.05
VS/TS (%)	70 ± 2	75 ± 1	82 ± 2
sCOD (mg/L)	1465 ± 45	2820 ± 52	850 ± 14
TVFA (mg/L)	1069 ± 10	2547 ± 15	124 ± 6
Alkalinity (mg/L)	8250 ± 11	469 ± 5	425 ± 4

adjusted to 7.5. The produced solution was incubated at 40 °C for 10 min. The reaction was stopped by adding 2.5 mL of TCA reagent. The samples were then passed through a 0.45 μ filter and 2 mL of the filtered solution was mixed with 5 mL of sodium carbonate ($NaHCO_3$) and 1 mL of Folin reagent. Finally, the solution was incubated at 37 °C for 30 min to measure the absorbance of different GE samples by spectrophotometry (DR-6000) at 660 nm against the blank. Detailed information on standard curve and color spectrum of different samples are provided in the supporting information (Table A.2 and Fig. A.2). Finally, the protease released from L-tyrosine and protease activity (U/mg) were calculated as:

$$\text{Protease activate (u/mL)} = \frac{(\text{released L} - \text{tyrosine (μmol)} \times \text{total volume (mL)})}{\text{enzyme solution volume} \times \text{color measurement solution volume} \times \text{time}} \quad (2)$$

2.3.3. Lipase activity

The lipase activity of GE was determined by the titration [30] for the reaction mixture containing 10 mL of pure olive live (10 % w/v), 5 mL arabic gum (acacia gum) (5 % *w/v*), 100 mL sodium hydrogen phosphate (0.1 M) and 100 mL distilled water that were mixed for 3 min. Next, 1 mL of GE solution was added to the produced solution and mixed again at 160 rpm 35 °C for 15 min. Next, 20 mL ethanol (C_2H_6O) and acetone (C_3H_6O) were added to the mixture at a ratio of 1:1 and finally titrated against 0.05 M NaOH until reaching pH of 9.5;p the amount of NaOH consumed was considered equal to the amount of lipase activity.

2.4. MA pre-treatment process

In this study, a combined MA pretreatment was used to measure the effect of air flow rate (AFR) and time on enhanced sludge solubilization and biogas production. A 500-mL batch glass reactor with a working volume of 400 mL, equipped with aquarium pump (Air-pump AC 9906) was used to determine the role of influencing factors on sludge sCOD. On the other hand, the roles of different levels of AFR (air volume in sludge volume in 1 min (vvm)) (0.1, 0.2, 0.3 and 0.4 (vvm)) and time periods (0, 20, 30, 40, 48, 60 h) on sCOD were investigated at 35 °C. Schematic diagram of MA reactors are presented in Fig. 1. In addition, ORP and pH were controlled by WTW inoLab pH 720, Xylem- WTW, Germany during the MA pretreatment.

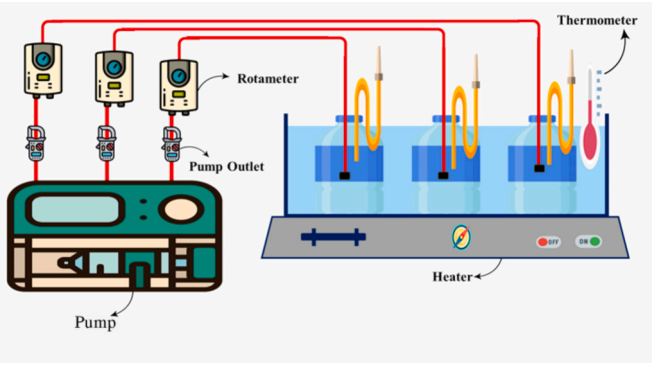


Fig. 1. A schematic diagram of lab-scale batch mode anaerobic digestion reactor.

2.5. Anaerobic reactors

2.5.1. Effects of different proportional ratios of GE

The experimental batch assays were performed to ascertain the cumulative methane production of sludge with different volume ratios of GE (5 %, 10 %, and 15 %) in anaerobic reactors and controls. In total, four plexiglass anaerobic reactors with total and working volumes of 0.5 and 0.4 L, placed in a mesophile incubators (37 ± 1 °C) were used to determine the effects of GE addition on BMP in anaerobic digestion process.

2.5.2. Effect of combined pretreatment method (MA + GE) on BMP

Considering the optimal proportional ratio of GE addition and MA, experimental batch assays were performed to ascertain biochemical methane potential (BMP) of the sludge pretreated with GE and MA in a separate and combined way. In total, five plexiglass anaerobic reactors with total and working volumes of 1 and 0.5 L, placed in a mesophilic incubator (37 ± 1 °C) were used to determine the effects of combined GE + MA on BMP in anaerobic digestion process. The information on different AD pretreated with different pretreatment methods and combined pretreatment is summarized in Table 2.

To operate the AD reactors, inoculum with a constant volume ratio (10 %) of working volume were added to each reactor. The co-substrates viz., primary and excess (30:70) sludge with predefined volume ratio were fed to the anaerobic reactors. Prior to incubation period (24 days), the batch reactors were purged with pure nitrogen gas for 5 min and sealed with gas-tight rubber stoppers to ensure anaerobic conditions. The stable mesophilic temperature was maintained by circulating hot water and controlled by the sensor. In addition, mixing was performed with a mechanical stirrer with a rotational speed of 60 rpm with 5-min on and 1-min off throughout the experiment. The volume of methane gas produced by the anaerobic reactors were determined by the water displacement method. The schematic of laboratory installation is presented in Fig. 2. The other operational parameters including soluble COD (sCOD), TVFA, alkalinity, DS, VS, and pH were analyzed in the predefined days during the incubation period. All the measurements were performed in triplicate.

2.6. Anaerobic digestion energy assessment

The energy balance of different AD including energy consumption (heating and mixing) and energy production (biogas) were estimated using Eqs. (3–5). The amount of electricity (in kJ/g VS removed) needed for mixing of AD were calculated using Eq. (3):

Table 2
Experimental setup of different anaerobic co-digested batch reactors.

Composition	Control	Reactors pretreated			
	R1	R2	R3	R4	R5
Pretreatment	–	MA	GE	GE + MA (separate)	GE + MA) simultaneous(
GE (mL)	0	0	36	36	36
PS (mL)	108	108	97.2	97.2	97.2
WAS (mL)	252	252	226.8	226.8	226.8
Inoculum (mL)	40	40	40	40	40
Working volume (mL)	400	400	400	400	400

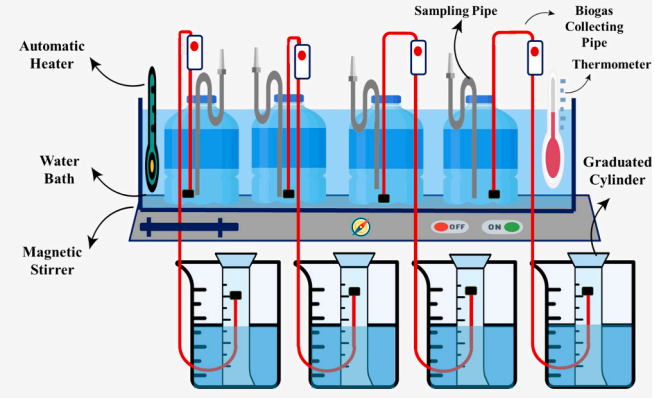


Fig. 2. Schematic diagram of lab-scale batch micro-aerobic pre-treatment.

$$E_{i.electricity} = \frac{V*300*24}{VS\ removed} \quad (3)$$

Here, the variable E_i represents the amount of electricity required for mixing (measured in kJ/g VS removed), V is the working volume (m^3), and VS removed refers to the volatile solids (g) consumed during the incubation period [31,32]. In addition, the energy requirement for heating and raising the temperature of AD were estimated according to Eq. (4):

$$E_{r.electricity} = \frac{\rho*V*\gamma*24*0.8}{VS\ removed} \quad (4)$$

where the variable Er is the quantity of electricity required (kJ/g VS removed), ρ stands for density of the influent (in units of 1000 kg/m^3), γ is specific heat of the influent (4.18 kJ/kg °C), V refers to the working volume in cubic meters, and VS removed represents grams of volatile solids consumed during the incubation period [31,32]. Furthermore, Eq. (5) was employed to calculate the energy generation based on methane production during the incubation period:

$$E_0 = \frac{P_{CH_4}*\varepsilon*\lambda_m}{VS\ removed} \quad (5)$$

where E_0 represents the energy output (measured in kJ/g VS removed),), P_{CH_4} stands for the cumulative methane production during the incubation period (in units of m^3), ε denotes the lower heating value of methane (35,800 kJ/m^3CH^4), λ_m represents the energy conversion factor of methane (0.9), and VS removed refers to the grams of volatile solids removed during the incubation period [33,34].

3. Results and discussion

3.1. Characterization of substrates and inoculum

Physicochemical characteristics of the primary, secondary sludge and inoculum samples summarized in Table 1 suggest that VS/ TS ratio in the primary and secondary sludge and inoculum were 75 ± 1, 82 ± 2 and 70 ± 2, respectively. The high ratio of VS to TS in primary and secondary sludge indicated that organic matters in sludge samples facilitate the biogas production from anaerobic digestion. In addition, the high VS/TS for inoculum (70 %) elucidate the presence of high amount of potential organic matters for anaerobic digestion and biogas production.

Table 3
Hydrolytic enzyme activity in different garbage enzyme solutions with variable pH.

Enzyme	pH	Concentration (μmol/mL)	Enzyme activity (U/mL)
Amylase	3.4	4.92	78.73
	5.5	5.00	80.08
	7.5	4.91	78.69
Protease	7.5	0.39	0.94
Lipase	9.5	–	44

3.2. GE characteristics (amylase, protease and lipase)

The specifications (sCOD and pH) of pure enzymatic solution extracted from the fermentation process of organic waste after 3 weeks were determined. The sCOD and pH of enzyme solution after the fermentation process were 4520 mg/L and 3.4, respectively. The enzymatic activity of GE solution containing amylase, protease and lipase are summarized in Table 3.

In addition, the values of different hydrolytic enzyme depending on different pH values are shown in Table 3 wherein the highest amylase activity (80.0894 U/mL) was determined in 5.5; the lowest amylase activity (78.69524 U/mL) was determined in the pH of 7.5. The reason for lower amount of α-amylase in other pH vakues is the change in 3D structure of the proteins [35,36]. [27] focused on the hydrolytic activity of GE extracted from the mixed organic waste pineapple, orange, tomato, cauliflower and mango, who reported that the increased pH led to increased activity of the amylase in GE, which is in consistence with the present results. In addition, comparing the activity of organic waste in the study of [37] showed that the activity values of enzymes, especially for amylase in orange wastes were higher than those of the other wastes. Higher enzyme activity for the waste enzyme produced from the oranges could be caused by high citric acid in the composition of this fruit. Extracellular polymeric substances (ESP) can be degraded by citric acids and release of hydrolytic enzymes [37].

In addition, the protease activity in GE were determined to be 0.940317 U/mL (see Table 3). The results indicated that the protease activity of this study was higher than the protease activity of the fermentation process of orange (0.38 U/mL) incubated in 90 days [27]. The most possible reasons for higher protease activity in the present study are attributed to higher operational temperature (35 °C) and microbial active area due to little size of orange peels. The lipase enzyme, which was measured by titration in pH 9.5 was calculated to be 44 mL. Rahman et al., [38] used the vegetable waste during the fermentation time of 30 days to extract the waste enzymes. The maximum activity of amylase, protease and lipase were measured to be 1.782 U/mg, 0.050 U/mg and 0.355 U/mg, respectively under the optimal conditions. Notice that these values are lower than the values measured in the present study.

3.3. Pretreatment processes

3.3.1. The effects of GE addition on biogas production

Table A.3 shows the effects of GE addition with different volumetric ratios (5, 10, and 15 %) on biogas production in anaerobic reactors. It is observed that the highest cumulative methane production (8100 mL) was in the reactor pretreated with 10 % GE, followed by 5 % (7095 mL) and 15 % (6900 mL), respectively. In addition, there are significant differences between the methane production in different reactors pretreated with GE and control (p-value<0.05). Nazim et al., [39] investigated the treatment of gray water using different proportional volume ratios of GE (5, 10, 20, 50 %); these results showed that 10 % (V/V) of GE was the optimum condition for the removal of TDS (58 %), BOD5 (65 %) and COD (79 %). The further increased volume ratio of GE increases the activity of enzymes and lower the pH, which suppresses the activity of the enzymes.

3.3.2. Effects of MA pretreatment on sludge solubility

MA pretreatment method was used to enhance the biodegradability index of the sludge before feeding into the AD process and biogas production. To this end, the effects of influencing parameters including AFR

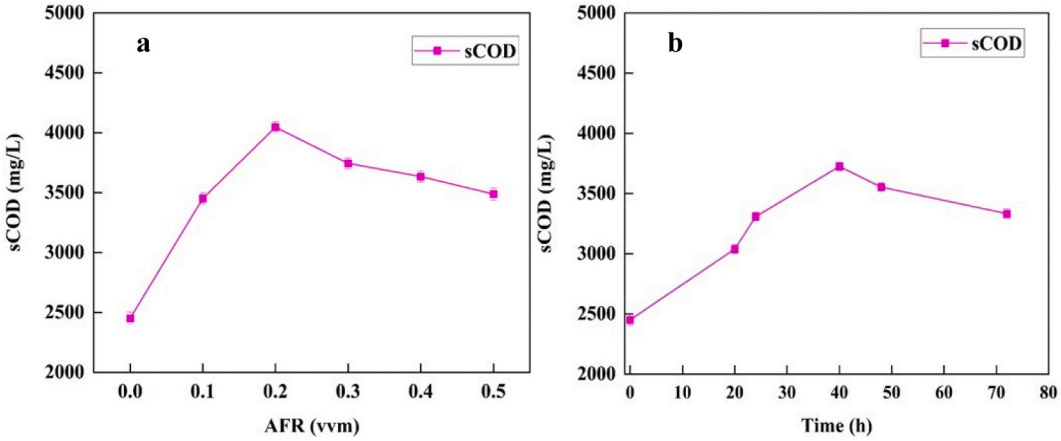


Fig. 3. Variation of sewage sludge sCOD in MA and effects of AFR (a) and time (b).

and time on sCOD was determined as shown in Fig. 3, which indicate the variation of sCOD as a function of AFR and time. As shown in Fig. 3(a), there is a two-stage trend for sCOD of sludge by increasing the AFR; the sCOD increased from 2450 mg/L to 4045 mg/L when the AFR increased from 0 in control sample to 0.2 vvm. However, the decreasing trend was observed by further increasing the AFR. Therefore, AFR equal to 0.2 vvm was considered optimum AFR with the highest sCOD.

In the case of time, there is a similar and two-step trend for sCOD of sludge (see Fig. 3 (b)). As shown in Fig. 3(b), the sCOD increased from 2450 mg/L to 3724 mg/L, when the time increased from 0 to 40 h. Therefore, the MA time equal to 40 h was considered as the optimum condition. The MA pretreatment increased the biodegradability and hydrolysis of the complex organic matters via providing suitable conditions for the growth of hydrolytic microorganisms and accordingly, the hydrolytic enzymes including the lipase [40,41]. Montalvo et al., [42] reported that the highest solubility of sewage sludge (PS: WAS: 40: 60) by the MA was achieved at 48 h of aeration and AFR equal to 0.3 vvm, which are comparable with the results presented here.

3.4. Anaerobic digestion performance

3.4.1. Biogas upgrading

The performance of anaerobic reactors in terms of daily biogas production and cumulative biogas production during the 24-day incubation period is presented in Fig. 4(a-b). The cumulative methane production from different reactors pretreated with separate, combined methods and control are presented in Fig. 4(a). As indicated in Fig. 4(a), the highest cumulative methane production was observed in R4 (9855 mL), where simultaneous MA and GE was performed for the pretreatment method, followed by R3 (9345 mL) with combined MA and GE pretreatment methods. Notice that R2 with MA pretreatment method and R1, as control produced 6105 mL and 5405 mL, respectively within the 24-day incubation period. These methane production analysis confirm that the simultaneous MA and GE pretreatment methods experienced synergetic effects on biodegradability of organic matters in anaerobic digestion, consequently increasing further the methane production. On the other hand, the simultaneous MA and GE increased more sCOD, as the biodegradability of sludge, thereby providing more substrate for methanogenic bacteria in order to produce more methane in the anaerobic digestion [43].

Fig. 4(b) shows the daily methane production in different anaerobic reactors, wherein similar trends are observed in different anaerobic reactors pretreated with different methods within the incubation period; the methane production increased from the initial days of the operation that gradually decreased until the end of the incubation period of anaerobic digestion. The highest methane production was observed in

Table 4
BMP from different anaerobic reactors; R1: control reactor; R2: microaerobic (MA); R3: combined (MA + GE); R4: simultaneous (MA + GE).

Parameters	Unit	R1	R2	R3	R4
Measured BMP	mLCH_4/g VS removed	344.92	390.09	580.07	609.46

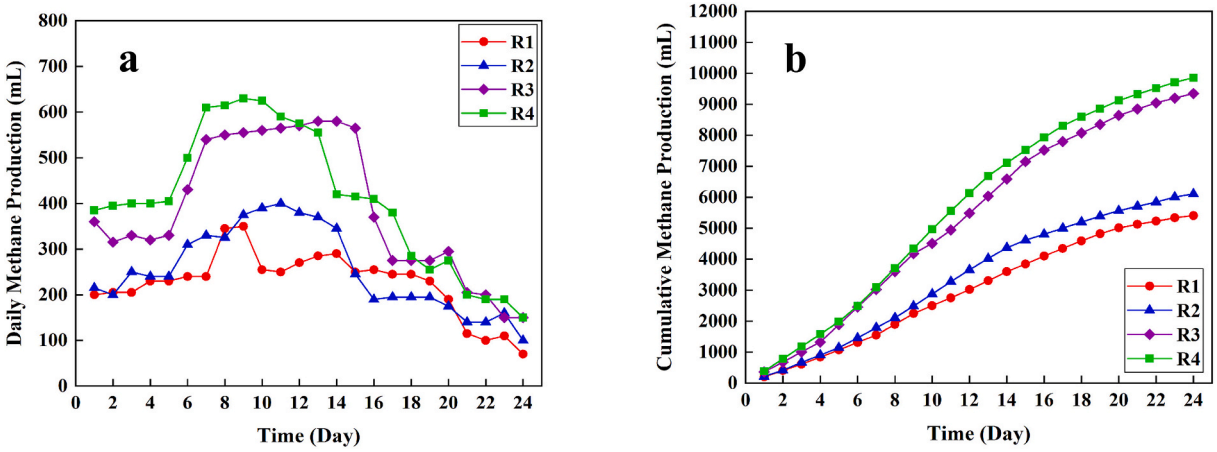


Fig. 4. (a-b). Process performance of anaerobic digested reactors in terms of daily biogas profile (a) and cumulative biogas production; (b) R1: control reactor; R2: microaerobic (MA); R3: combined (MA + GE); R4: simultaneous (MA + GE).

R4 (630 mL) in day 9, followed by R3 (580 mL) in day 13. There is a significant difference between the highest daily production between the reactors assisted with different pretreatement methods and control (350 mL) (p-value <0.05).

Table 4 summarizes detailed information on the BMP of different anaerobic reactors assisted with different pretreatment methods, where the highest BMP (609.46 mLCH_4/g VS removed) was observed in R4, in which the simultaneous GE and MA were employed to solubilize the sewage sludge, followed by R3 (580.07 mLCH_4/g VS) and R2 (390.09 mLCH_4/g VS removed), respectively. On the other hand, simulataneous GE and MA enhanced the BMP by 1.76 fold compared to the control. Zhang et al., [44] investigated the effects of microwave pretreatment (MW) on biogas upgrading of anaerobic reactor fed with food waste (FW) and sewage sludge (SS). The authors reported that the highest BMP (338.44 mL CH4/ g VSadded) when the optimal ratio of FW:SS (3:2) was fed to the mesophilica anaerobic reactors. Cherukuri et al., [45] focused on the improvement of the biogas production from anaerobic digestion of FW using the thermal pretreatment under mesophilic temperature. The highest BMP (730 mL/g VSfed) was achieved, which is comparable with the results of this study.

3.4.2. The sCOD variation and VS reduction (VSR) in anaerobic reactors

Fig. 5 shows the sCOD variations in different anaerobic digestion reactors, where a similar two-step trends for sCOD were observed in different anaerobic reactors during the incubation period. The sCOD increased during the fisrt and lag phase of anaerobic digestion, which gradually decreased due to the consumption of methanogenic bacteria as the substrates [46]. In addition, the highest increased sCOD concentration was observed in R4 (the difference between day 5 and day 1, was 2447 mg/ L), followed by R 3 (1773), R2 (270), and R1 (270). The simultaneous GE and MA pretreatments were able to effectively disintegrate the biological solids of the sludge to provide more soluble matters for microorganisms [47]. Arun et al., [18] focused on the synergic effects of ultrasound and GE on the waste activated sludge (WAS) of the dairy industry. The results showed that the addition of 35 mL of GE to 50 mL of WAS of dairy products experienced the greatest decrease in SS, sCOD, soluble protein, carbohydrates 27.45 %, 28.36 %, 987 mg/ L and 932 mg /L, respectively. Furthermore, at the end of the incubation period, the remaining sCOD in R4 and R3 were observed to be between 4200 and 4300 mg/L, while the remaining sCOD for R2 (pretreated by MA) and control were and 1200 and 1110 mg/L, respectively.

The VSR for different anaerobic reactors within the incubation period are summarized in Table 5, where the highest VSR was observed in R3 (52.07 %); here, the combined GE + MA was used to enhance the methane production followed by R4 (49.02 %) and R2 (41.45 %).

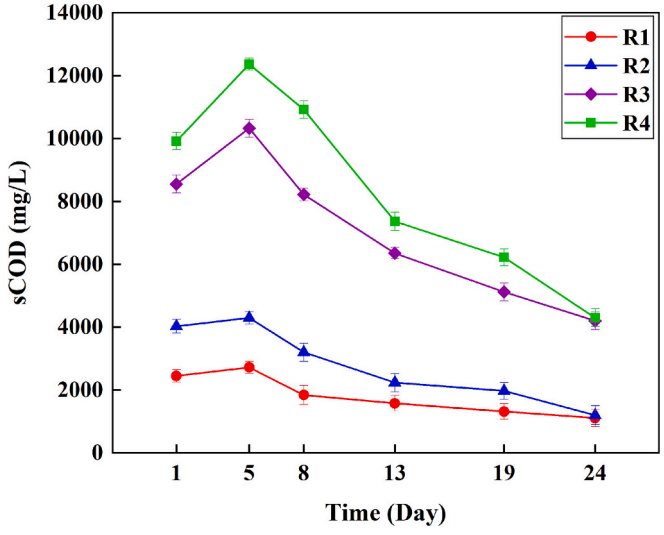


Fig. 5. Dynamics of sCOD in different anaerobic digester reactors; R1: control reactor; R2: microaerobic (MA); R3: combined (MA + GE); R4: simultaneous (MA + GE).

Table 5
Initial and final fraction of volatile in anaerobic pretreated GE and MA reactors and control.

Parameter	R1	R2	R3	R4
Initial VS/DS (%)	71	64	71	68
Final VS/DS (%)	61	51	54	52
VSR (%)	36.11	41.45	52.05	49.02

3.4.3. pH and total volatile fatty acids (TVFA) variations in anaerobic reactors

The pH variation of different anaerobic reactors illustrated in Fig. 6 (a), suggest the same trend of pH for different anaerobic reactors; the pH showed decreasing trend in initial and hydrolysis stages of the anaerobic digestion that gradually increased when the methanogenic bacteria consumed TVFA for methane production [48]. The lowest pH was observed in R4 (5.9), where simultaneous GE and MA were used for enhanced methane production. The higher TVFA in R4 and R3 are the main possible reasons for lower pH in initial stages of the anaerobic digestion compared to other reactors. In addition, the same trend of pH variations were observed in different studies focused on the improvement of biogas production with pretreatment methods [49,50].

The TVFA variations in different anaerobic reactors illustrated in Fig. 6(b) shows that TVFA in different reactors increased during the first phase of anaerobic reactors, which gradually decreased due to the consumption of TVFA for methane production by the methanogenic bacteria. The highest TVFA was observed in R4 (6050 mg/L) followed by R3 (5120 mg/L). The results indicate that simultaneous pretreatment methods (GE + MA) significantly increased the TVFA needed for methanogenic bacteria in order to produce more of methane [51,52]. Lu et al., [53] surveyed the VFA production from the potato peel waste. The authors examined the effects of different pH levels (5, 7, 11) on the concentration and composition of VFA, intermediate products and metabolic status, showed the highest TVFA, suggesting that the highest concentration of total VFA was 41.9 mg COD/L obtained at pH 7.0.

3.5. Potential energy recovery from anaerobic co-digestion

Table 6 summaries the energy balance including energy production and energy consumption (electricity and heat) for different anaerobic reactors assisted with different pretreatment processes. The output energy obtained from the anaerobic reactors were estimated to and found to be between 108.36 and 193.70 kJ/g VSremoved; the highest value was for R4 (193.70 kJ/g VSremoved), followed by R3 (184.15 kJ/g VSremoved). Therefore, the energy yield in pretreated anaerobic reactors with MA combined and simultaneous pretreatment experienced increases of 14.51 kJ/g VS, 75.71 kJ/g VS and 85.45 kJ/g VS compared to control reactors without the pretreatment process.

Additionally, the total energy balance, difference between the energy yield and energy consumption by different anaerobic reactors indicated that there is a significant difference between the energy balance for reactors assisted with pretreatment methods and control; the highest energy balance observed was for R4 (193.70 kJ/g VS). This differences was estimated to be 85.34 kJ/g VS compared to the control (108.36 kJ/g VS). Liu et al., [54] studied the effect of microwave (MW) pretreatment on the anaerobic digestion of sludge (SS) and food waste (FW). The results obtained showed the highest net energy output achieved using the MW pretreatment that were 76.25 kJ/g Fed VS.

4. Conclusions

The present study investigates the SS solubilization and

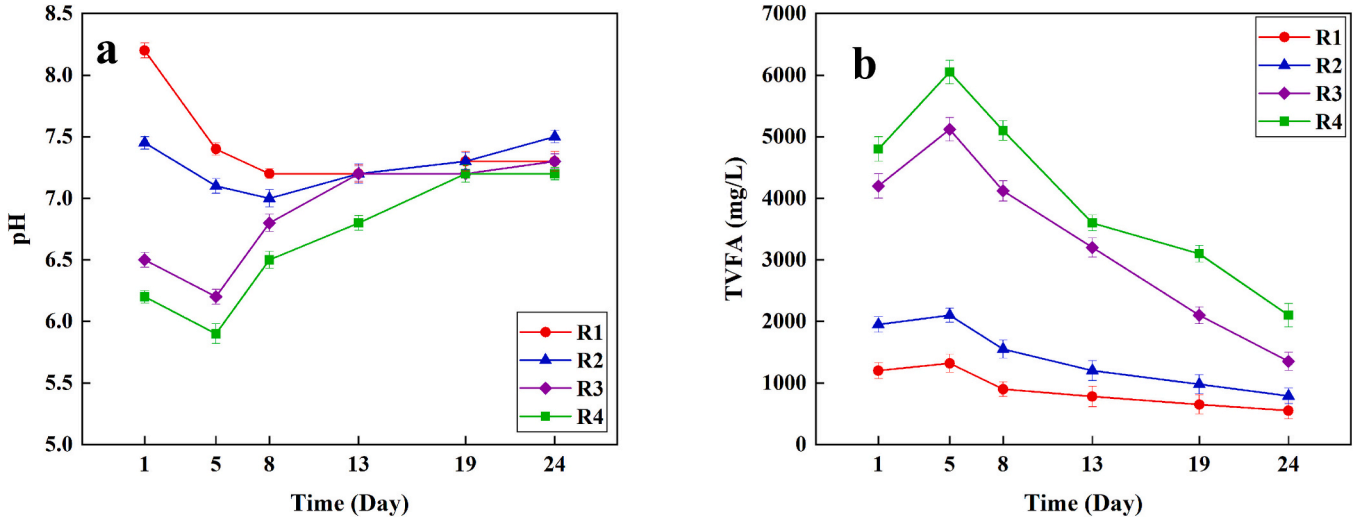


Fig. 6. Dynamics of the changes in pH (a) and TVFA (b) over time in the different anaerobic digestion reactors; R1: control reactor; R2: microaerobic (MA); R3: combined (MA + GE); R4: simultaneous (MA + GE).

Table 6
Energy assessment in different anaerobic digester reactors; R1: control reactor; R2: microaerobic (MA); R3: combined (MA + GE); R4: simultaneous (MA + GE).

	R1	R2	R3	R4
Energy output (kJ/g VS removed)	111.15	125.66	186.86	196.40
Input electricity(kJ/g VS removed)	0.23	0.23	0.22	0.22
Input heat (kJ/g VS removed)	2.56	2.56	2.49	2.48
Energy balance (kJ/g VS removed)	108.36	122.87	184.15	193.70

bioavailability for enhanced methane and energy production using eco-friendly and economic pretreatment method. The GE fermented from GE and MA were utilized to enhance the energy production from SS. It was found that simultaneous GE and MA is a feasible promising technology to enhance methane and energy production from the SS under mesophilic temperature. The highest BMP (609.46 mLCH_4/g VS removed) were achieved in R4, where the simultaneous GE and MA were utilized under the optimal conditions. The highest cumulative methane production (9855 mL) was observed in R4, which was 1.82 folds higher than the control (5405 mL). The energy balance analysis indicated a significant difference between the highest energy yield (+193.70 kJ/g VS) and the energy produced from the control (+108.36 kJ/g VS). Therefore, the SS anaerobic reactor can be considered as the promising and cost-effective method. Further research is recommended to investigate the methane content and methanogenic pathway occurring by the predominant microorganisms. The present study meets the required criteria of UNSDGs Program #SGD6, #SDG7, #SDG12 and #SGD13.

CRediT authorship contribution statement

Somayeh Moradi: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Hasan Pasalari:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis. **Roya Sharifi:** Visualization, Software, Resources, Conceptualization. **Mitra Gholami:** Visualization, Software, Resources, Formal analysis. **Mahdi Farzadkia:** Writing – review & editing, Validation, Resources, Investigation. **Sami Rtimi:** Validation, Supervision, Resources. **Tejraj M. Aminabhavi:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The present project was supported by Iran University of Medical Sciences [Grant number 1401-4-2-25134].

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2025.168425.

Data availability

Data will be made available on request.

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